Ryan Hartung

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Education

The University of Notre Dame — Notre Dame, IN
Ph.D. in Computer Science & Engineering — *Expected May 2029*
- Advisor: Dr. Douglas Thain
- Research Focus: Distributed Systems and High-Performance Computing

The University of Notre Dame — Notre Dame, IN
Master of Science in Computer Science & Engineering — *August 2025*

The University of Texas at Austin — Austin, TX
Bachelor of Science in Physics (Computation Option) — *May 2024*
- Certificate in Elements of Computing

Publications

[2026] Liubov Kurafeeva, **Ryan Hartung**, Benjamin Carter, Alan Subedi, Avhishek Biswas, Michael Fay, Shantenu Jha, Chandra Krintz, Andre Merzky, Douglas Thain, Mehmet Vuran, and Rich Wolski. "Hybrid Edge-HPC Systems for Low-Latency Data-Driven Inference." *arXiv preprint arXiv:2605.20532*. https://arxiv.org/abs/2605.20532

[2025] Liubov Kurafeeva, Alan Subedi, **Ryan Hartung**, Michael Fay, Avhishek Biswas, Shantenu Jha, Ozgur Kilic, Chandra Krintz, Andre Merzky, Douglas Thain, Mehmet Vuran, and Rich Wolski. "xGFabric: Coupling Sensor Networks and HPC Facilities with 5G Wireless Networks for Real-Time Digital Agriculture." In *Proceedings of the SC '25 Workshops*. ACM, 2317–2327. DOI: https://doi.org/10.1145/3731599.3767589

Research Experience

The University of Notre Dame — Notre Dame, IN
Graduate Research Assistant — January 2025 – Present
- Contributing to the design and implementation of xGFabric, an architectural framework connecting remote sensor edge networks to HPC centers via 5G.
- Developing low-latency data pipelines to facilitate real-time inference across heterogeneous edge-HPC infrastructure.

The University of Texas at Austin — Austin, TX
Undergraduate Research Assistant — June 2022 – December 2023
- Engineered an open-source Python tool analyzing star-spot patterns across 6,000+ star systems with optimized array processing algorithms.
- Evaluated star-spot contrast ratios using observational data from NASA Kepler and TESS missions to establish baseline metrics for exoplanet detection.
- Integrated parallel processing and multi-core compilation to optimize data throughput and execution runtime.
- Published codebase and workflows to an open-source GitHub repository; research funded by NASA grant.

Work Experience

Hargrove Engineers and Constructors — Katy, TX
Controls & Automations Engineering Intern — May 2024 – August 2024
- Automated the migration of 200,000+ lines of Dow MOD5 code into Honeywell Experion with no loss of functionality.
- Tracked, analyzed, and documented over 7,000 variables to design flow diagrams visualizing variable definitions.
- Developed several algorithms to expedite the migration and variable mapping processes, improving overall efficiency.

Hargrove Engineers and Constructors — Katy, TX
Controls & Automations Engineering Intern — May 2023 – August 2023
- Instantiated existing library modules to complete Westlake's PVC plant Reactor and Train configuration.
- Performed safety testing on all module and sequence configurations on simulated ABB hardware.
- Successfully configured Modbus communication between the ABB and HIMA system controllers.
- Updated structured text configuration to handle cross-controller communications.

Talks, Panels, & Presentations

GCASR 2026 Evanston, IL
Poster Presenter May 11, 2026

- Poster: *Hybrid Edge-HPC Systems for Low-Latency Data-Driven Inference*.

Supercomputing '25 St. Louis, MO
Presenter and Demonstrator November 17 – 19, 2025

- Presented xGFabric and its findings at the Department of Energy's (DOE) booth.
- Gave a live demo to DOE staff, SC attendees, and fellow researchers, showcasing:
  - Real-time data transfer between a Raspberry Pi and a 5G sensor, with live latency and throughput metrics.
  - The full end-to-end pipeline: fetching and parsing data, launching simulations, and rending results.

Supercomputing '25 St. Louis, MO
Invited Panelist November 16, 2025

- Participated in an expert panel on edge-to-HPC loop architectures and the future of computing during the XLOOP Workshop [Recording available online].

HPDC 2025 Notre Dame, IN
Poster Presenter July 21, 2025

- Poster: *xGFabric: Coupling Sensor Networks and HPC Facilities with 5G Wireless Networks for Real-Time Digital Agriculture*.

Teaching Experience

The University of Notre Dame Notre Dame, IN
Graduate Teaching Assistant August 2024 – May 2025

- CSE 10102; Elements of Computing II (Spring 2025): 37 students; held office hours, graded, assisted with in-class coding.
- CSE 10001; Principles of Computing (Fall 2024): 35 students; proctored exams and graded.

Technical Skills

- **Programming Languages:** Bash, C, Python, Rust, Java, JavaScript/NodeJS, SQL
- **Web Development:** CSS, Django, HTML, Jekyll, React
- **Batch Schedulers:** HTCondor, SLURM, Univa Grid Engine (UGE)
- **Tools & Platforms:** ABB, Docker, Git, Honeywell, LaTeX, Linux/UNIX, MPI, OpenFOAM, OpenMP, ParaView

Leadership & Service

Order of Christian Initiation of Adults (OCIA), Sponsor — September 2026 – Present
Fischer Graduate Residence Mass, Lector — August 2025 – May 2026
Society of Catholic Scientists, Member — August 2025 – Present
Science Policy Initiative at Notre Dame, Member — August 2025 – Present
Catholic Graduate Community, Member — April 2025 – Present
Technology and Catholicism Club, Member — February 2025 – Present
Notre Dame Rock Climbing Club, Member — August 2024 – Present
Notre Dame Texas Club, Member — August 2024 – Present
Science Olympiad, Competitor, Mentor, President, Event Supervisor, and Lab Coordinator — August 2011 – Present